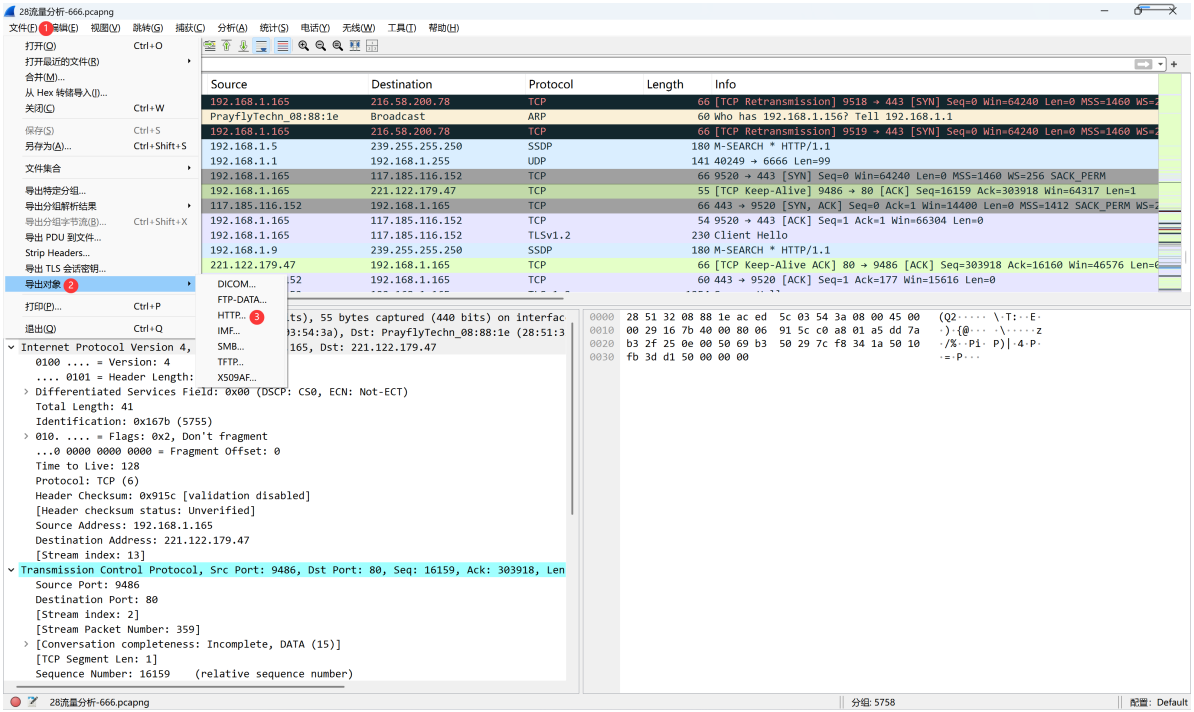
进制转换

给了一串文本，根据各种进制的特征可以判断出 `o` 开头为 8 进制，`d` 开头为 10 进制，`x` 开头为 16 进制，`b` 开头为 2 进制。

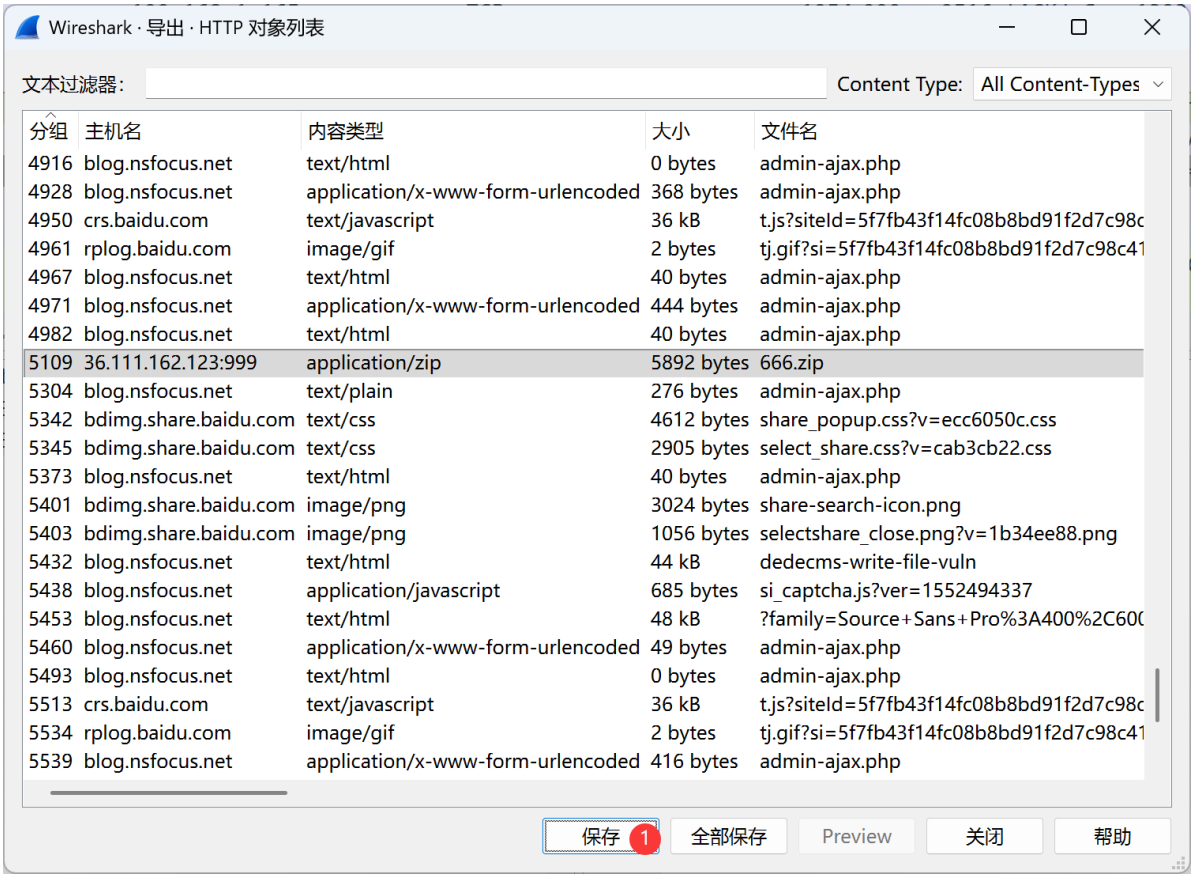
那么目前的思路是，分别在前面补 0 （10 进制除外），然后利用 int 函数的第二个参数来实现自动转 10 进制后转 Ascii 码。

```
s = 'o40 d71 x69 d118 x65 x20 b1111001 o157 b1110101 d32 o141 d32 d102 o154 x61 x67 b100000 o141 d115 b100000 b1100001 d32 x67 o151 x66 d116 b101110 b100000 d32 d102 d108 d97 o147 d123 x31 b1100101 b110100 d98 d102 b111000 d49 b1100001 d54 b110011 x39 o64 o144 o145 d53 x61 b1100010 b1100011 o60 d48 o65 b1100001 x63 b110110 d101 o63 b111001 d97 d51 o70 d55 b1100010 d125 x20 b101110 x20 b1001000 d97 d118 o145 x20 d97 o40 d103 d111 d111 x64 d32 o164 b1101001 x6d o145 x7e'

s = s.replace('o', '0o').replace('d', '').replace('x', '0x').replace('b', '0b').split()

flag = ''
for i in s:
    flag += chr(int(i, 0))
print(flag)
#  flag{1e4bf81a6394de5abc005ac6e39a387b}
```

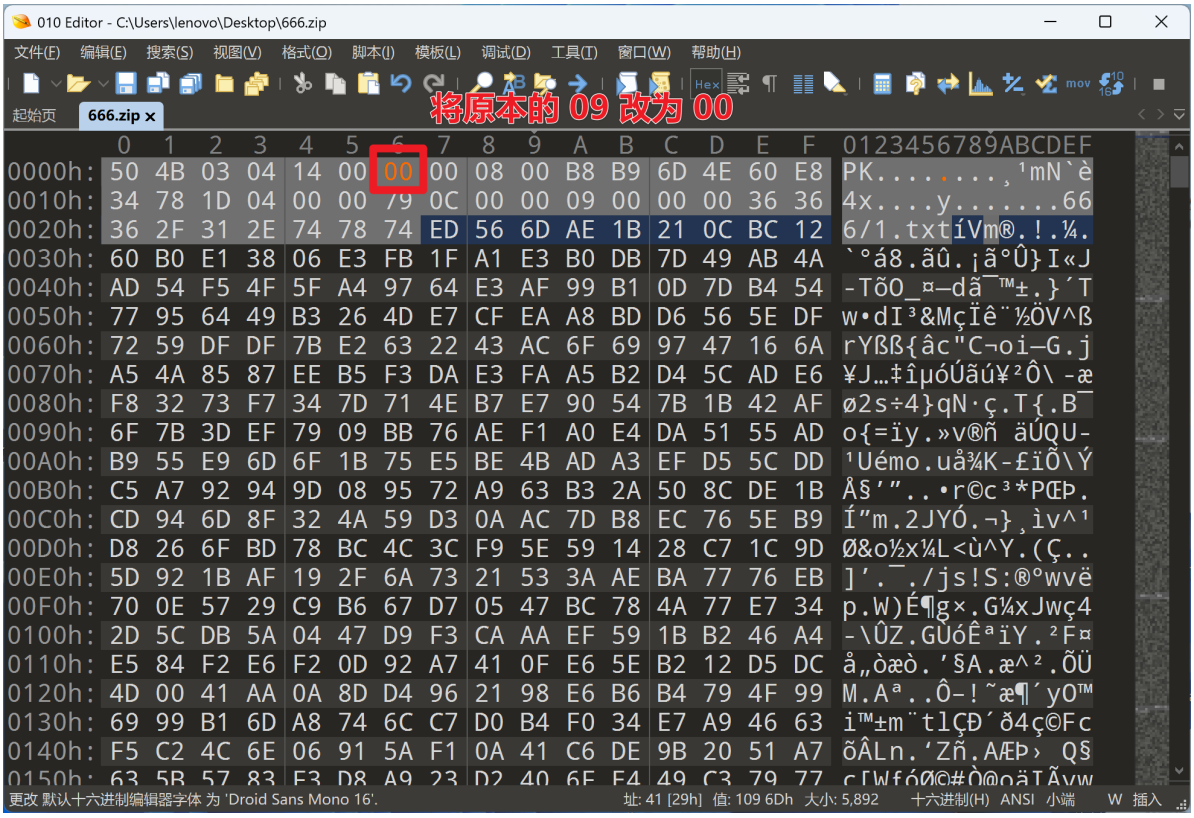
流量分析-666



在弹出的 `HTTP 对象列表` 中发现 `666.zip`



导出后提示输入密码，使用 `010editor` 打开后判断存在伪加密



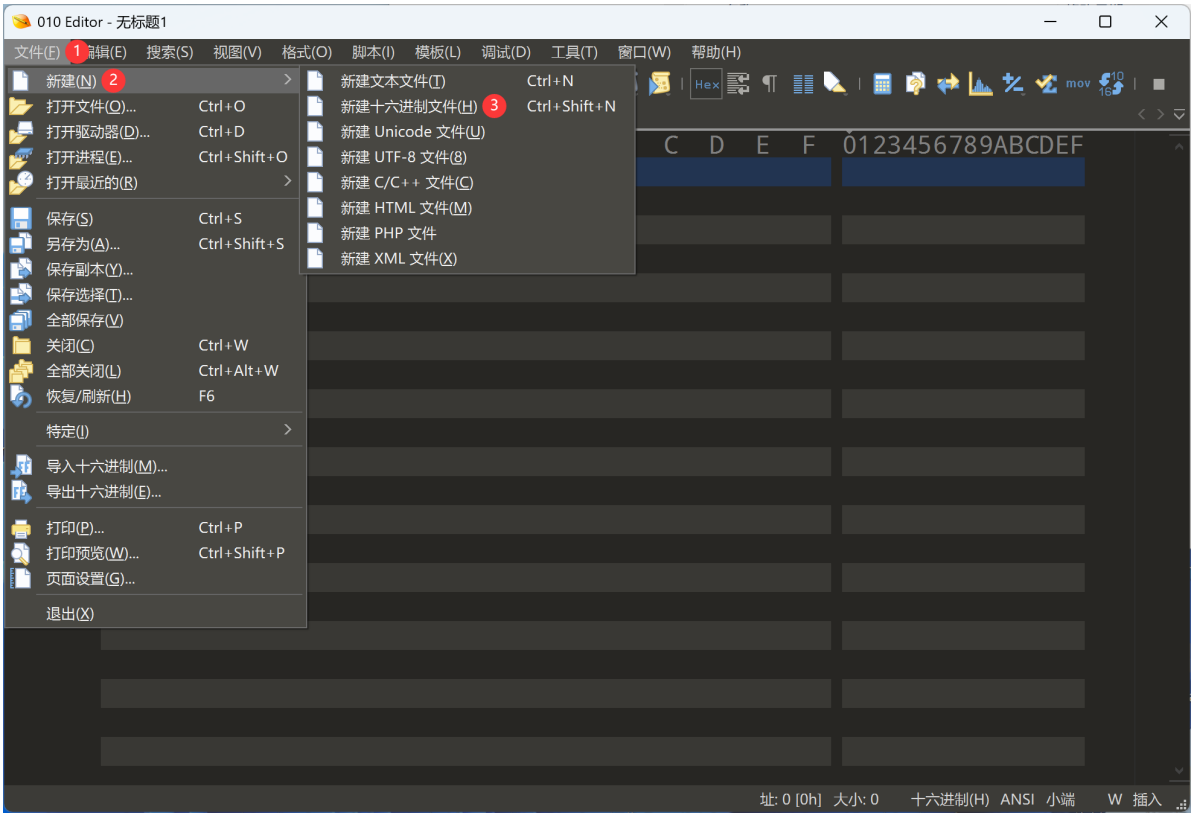
修改加密位后解压，解压后得到 6 个 `txt` 文件，其中 1.txt 开头为 `89504e47` ，那么逻辑清楚了，直接将 6 个 txt 依次进行拼接

注：下面的命令必须通过 cmd 执行

```
copy /b *.txt flag.txt
```

```
Windows PowerShell - powers
PS C:\Users\lenovo\Desktop\666> ls


    目录: C:\Users\lenovo\Desktop\666


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        2019-03-13     23:13           3193 1.txt
-a----        2019-03-13     23:14           2781 2.txt
-a----        2019-03-13     23:14           2987 3.txt
-a----        2019-03-13     23:14           3090 4.txt
-a----        2019-03-13     23:15           2987 5.txt
-a----        2019-03-13     23:15           2218 6.txt


PS C:\Users\lenovo\Desktop\666> cmd
Microsoft Windows [版本 10.0.26200.6899]
(c) Microsoft Corporation。保留所有权利。

C:\Users\lenovo\Desktop\666>copy /b *.txt flag.txt
1.txt
2.txt
3.txt
4.txt
5.txt
6.txt
已复制         1 个文件。

C:\Users\lenovo\Desktop\666>
```

然后再 `010editor` 中新建十六文件，将处理后的 `flag.txt` 复制后使用 `Ctrl+shift+V` 粘贴至左侧 HEX 区域，使用 `Ctrl+S` 将其保存到 `flag.png` 中



保存后发现 `flag.png` 为一张二维码图片，使用 `CQR` 软件对其进行扫描后得到 flag

```
flag{3819169573b7a37786d2ea39c6daef76}
```

真的RSA

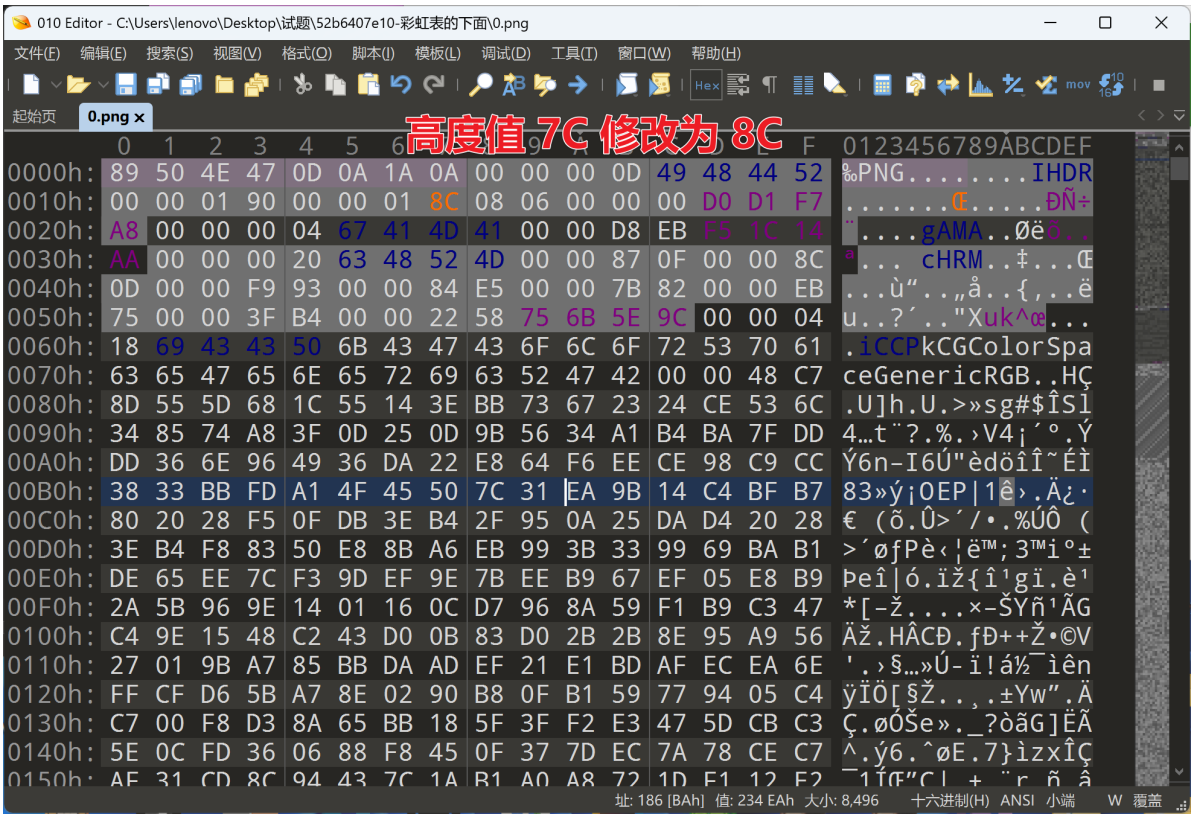
已知 n、e、c，通过 factordb 对 n 进行因式分解得到 p 和 q 后直接解即可。

```
from libnum import n2s

p =
159918469709932133220726269015607499326863257664034048640233418107353192490663709160906409262190793688455104440314003222291477716829611324204818973628431 99
q =
288057917712602594868569027290204386866703544412962471482078628360646578497353436182070981639017872873685697684725213446355673342993567600805074546402070 03
```

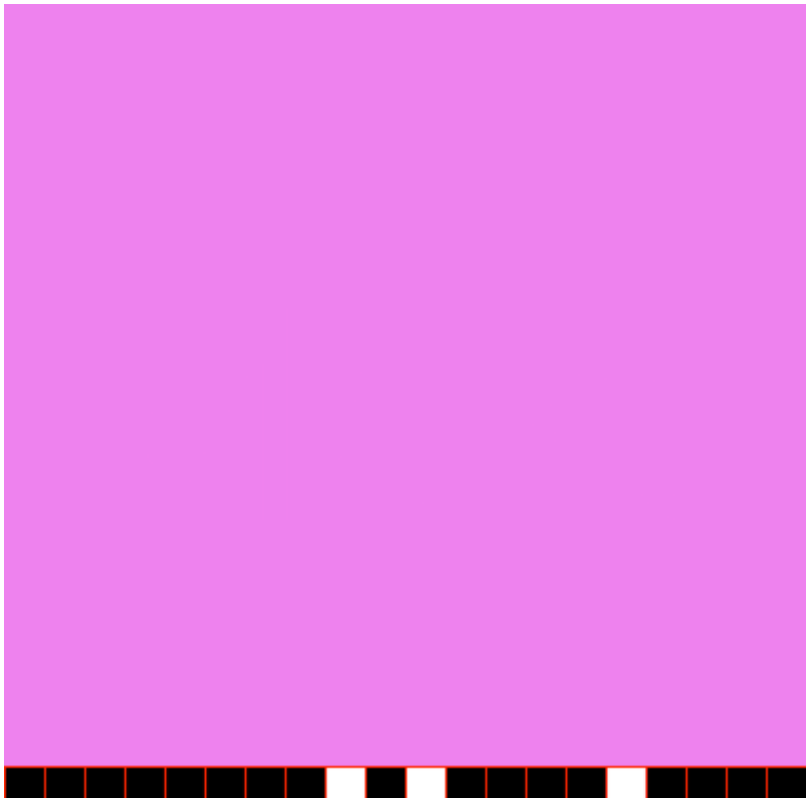
```
e =
35461110244130757205657218182792589919834535022875373093108939327546391654445662
68942454150961078344657784095323731871253185546147225993017915289162128393681210
66035541008808261534500586023652767712271625785204280964688004680328300124849680
47710530251937737009257810782711682139182621097232037761496754782761 9
enc =
38230991316229399651823567590692301060044620412191737764632384680546256228451518
23884296522139471184833783245944384444688946836215418821484073674465788585894381
01776758719911114666531582571911396056999163473082949956645302808168504827405306
02254559123759121106338359220242637775919026933563326069449424391192

phi = (p - 1) * (q - 1)
d = pow(e, -1, phi)  # 利用pow负指数直接求模逆
m = pow(enc, d, p * q)
print(n2s(m))
# flag{Wien3r_4tt@ck_1s_3AsY}
# 如果没有 libnum 可以考虑用 binascii
# print(binascii.unhexlify(hex(m)[2:]))
```

彩虹表的下面

下载附件得到 7 张 png 图片，结合题目名称经检测存在高度隐写



修改后可以看到图标底部出现一串黑白字符串

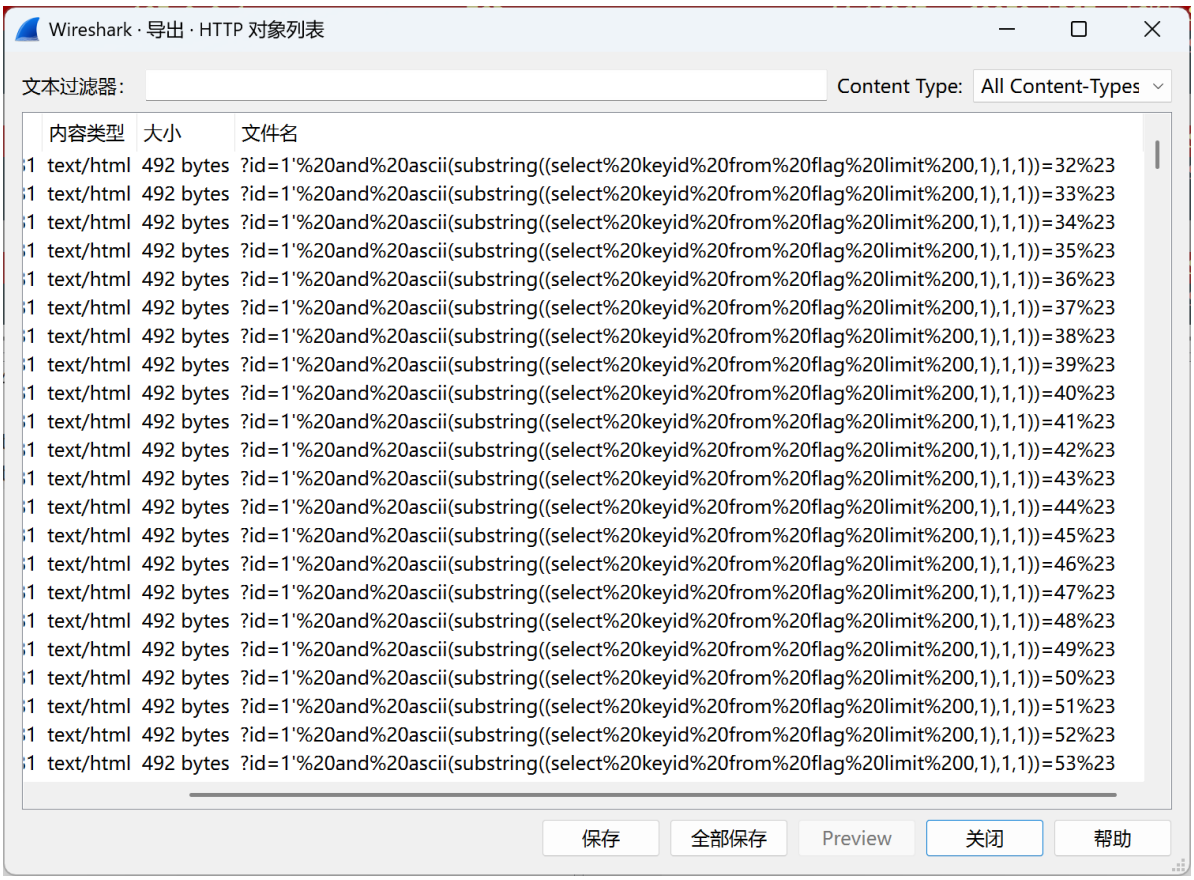


将每张图片的黑色设为 1，白色设为 0，然后直接竖着把二进制转字符串即可

```
c = [
    '11111111010111101111',
    '11111011111110111111',
    '00001100101010110001',
    '01001010010000001101',
    '11010011011101010111',
    '10011011011010110110',
    '00111001101101111101'
]

flag = ''

for i in range(20):
    s = ''
    for j in c:
        s += j[i]  # 拼接每个字符串的第i位

    flag += chr(int(s, 2))

print(flag)
# flag{Png1n7erEs7iof}
```

注意尾部

根据 流量分析-666 题目的步骤，再 `HTTP 对象列表` 中发现 SQL 盲注流量



那么我们直接利用 `tshark` 筛选出完整 URL 中包含字符串 flag 的 HTTP 请求。

```
tshark -r .\19-MISC1.pcapng  -Y 'http.request.full_uri contains "flag"' >
data.txt
```

然后直接利用代码自动取位后转 Ascii 码得到 flag

```python
import re

with open('data.txt', 'r', encoding='utf-16') as f:
    tmp = f.read()

    res = re.findall(r'\)\)=(\d+)%', tmp)
    res = [int(i) for i in res]

    flag = ''
    for i, num in enumerate(res):
        try:
            if num > res[i + 1]:
                flag += chr(num)
        except:
            pass

# 代码逻辑不会处理最后一个数字不，需要手动添加 }，或者输出前加上下面这段代码
# flag += chr(res[-1])
print(flag)
# flag{c2bbf9cecdaf656cf524d014c5bf046c}
```

怎么还是RSA

题目已知 e1、e2、c1 和 c2，那么肯定是共模攻击了。

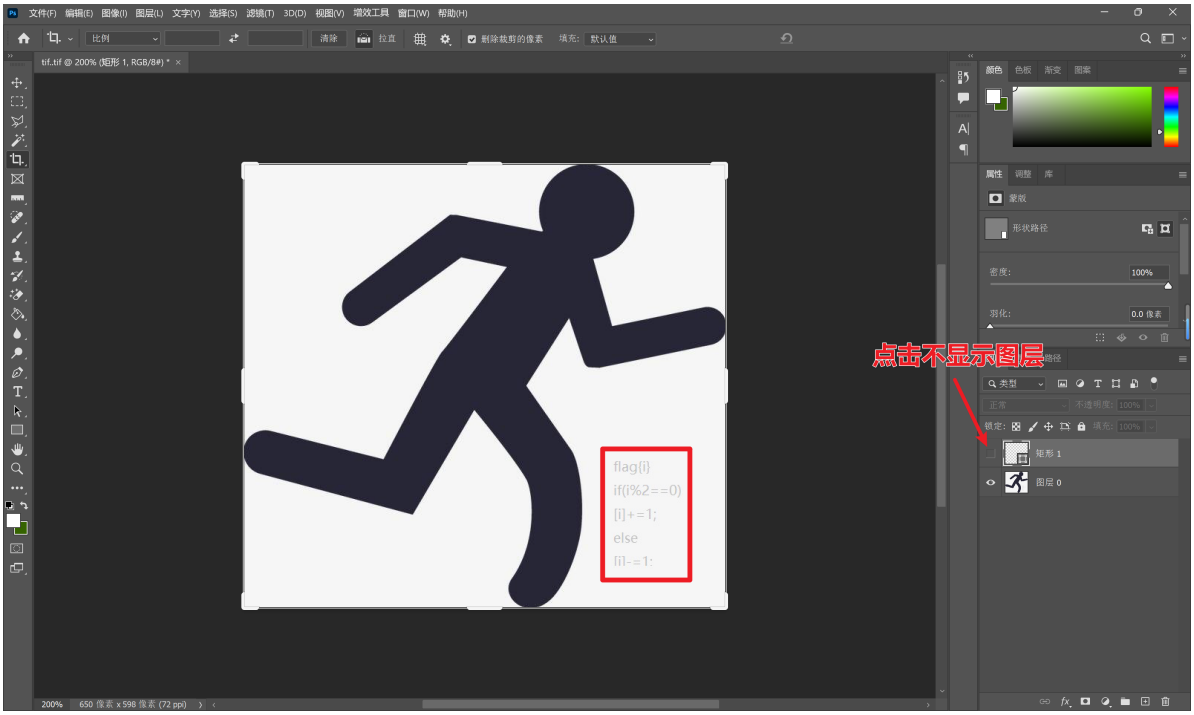
```python
from gmpy2 import invert, gcdext
from libnum import n2s

def gongmo(n, c1, c2, e1, e2):
    _, s1, s2 = gcdext(e1, e2)
    if s1 < 0:
        s1 = -s1
        c1 = invert(c1, n)
    if s2 < 0:
        s2 = -s2
        c2 = invert(c2, n)
    return (pow(c1, s1, n) * pow(c2, s2, n)) % n

c1=22322035275663237041646893770451933509324701913484303338076210603542612758956
2628696408224864701211494244855713610074212936755163388221952803137949911360481 4
0918842471219840263536338886250492682739436410013436651161720725855484866690084 7
8872134955566201987908150111322299612330553300932596437779889270316152185280595 6
8112195638833128963301562986216746843539195475581279209257068428089147621990110 5
4955816534977675267395009575347820387073483928425066536361482774892370969520740 3
0428745655550893337278232750656901077253749754176431142905221629119893209261779 2
6452539014789108015928782035648611189120454649598325660513 61
n=2270807881588501146246204906433918589871243927722683107345788840312937854735 02
9242026701655181905243077900475584664904400102414148528328648313070261605727469 8
4736111495087988697063475019315831176327107007872280164801276773936499295304165 9
8686027354216422565934459015161927613607902831542857977859612596282353679327773 3
0372700440726219723158632459918198357262240459035408454178806226216451014060586 8
1224103880901744201477524085541297897609023008980462739090078528184740307706996 4
7647363015102118956737673941354217692696044969695308506436573142565573487583507 0
3735694484803986438233921626667067356748887150892531115480 1
e1=11187289
c2=1870201004518701555654869164239498283566926214723021273130993867522645855521 0
4259724294184492734105353879859310367118542656239050668056657518032691068807467 6
9003478900791099590239513925449748814075904017471585572848473556490565450062664 7
0644912841583478796194726625978978596292223870113407972041422841406619307149530 4
6123410529874556159300235368238014992697733571860874527475008406404193650115544 2
1183037505653461286732740983702740822671148045619497667184586123657285604061875 6
5390956782232891406533779773344464035151877548764981997826236361726579798284317 9
6308887294072384966509877204287082171152579890078673316983 97
e2=9647291

flag = int(gongmo(n, c1, c2, e1, e2))

print(n2s(flag))
# flag{49d91077a1abcb14f1a9d546c80be9ef}
```

神秘的右下角

运行 `神秘的右下角.exe` 后，生成了 `tif` 文件，我们手动给其添加上 `.tif` 后缀后通过 Photoshop 打开

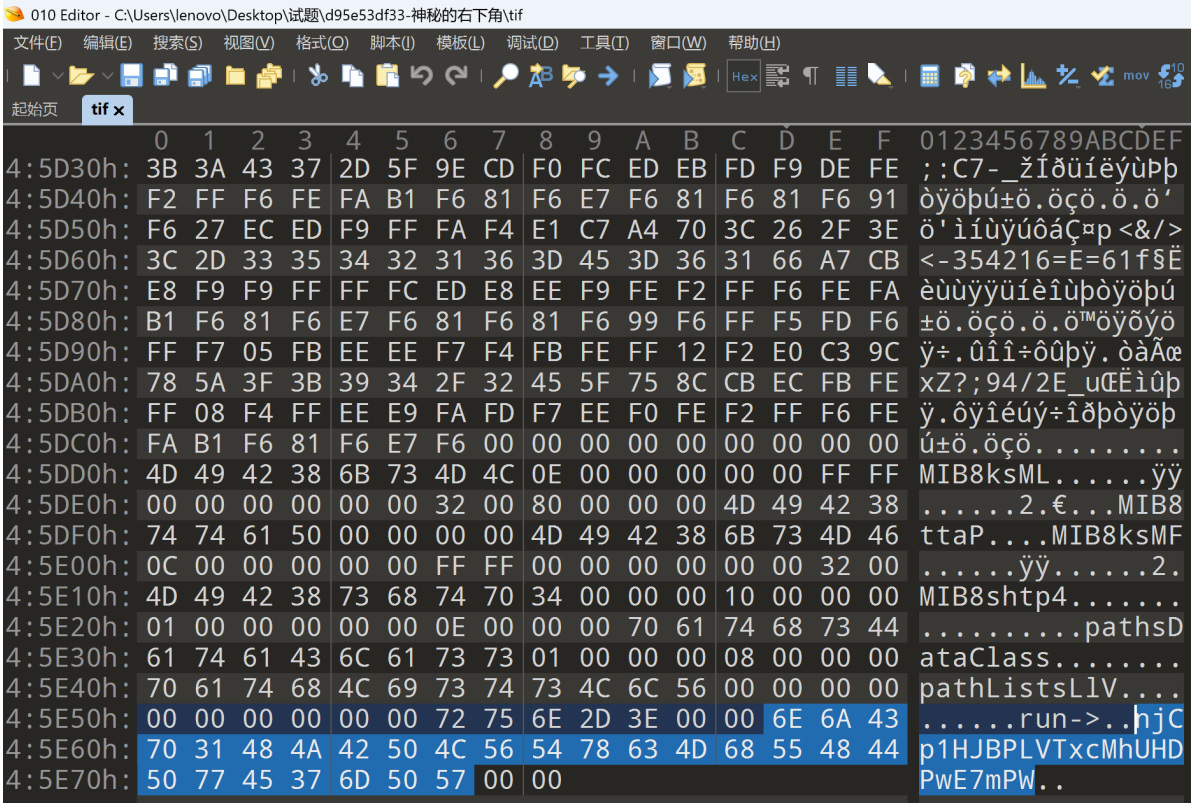
名称	日期	类型	大小	标记
33-神秘的右下角.exe	2020-06-17 15:45	应用程序	359 KB	
tif.tif	2020-06-17 15:26	TIF 文件	280 KB	

发现有一个黑块，隐藏黑块图层后得到一段代码



同时在 `010editor` 的尾部发现了一段密文，直接选中后 `右键` 复制即可

```
njCp1HJBPLVTxcMhUHDPwE7mPW
```

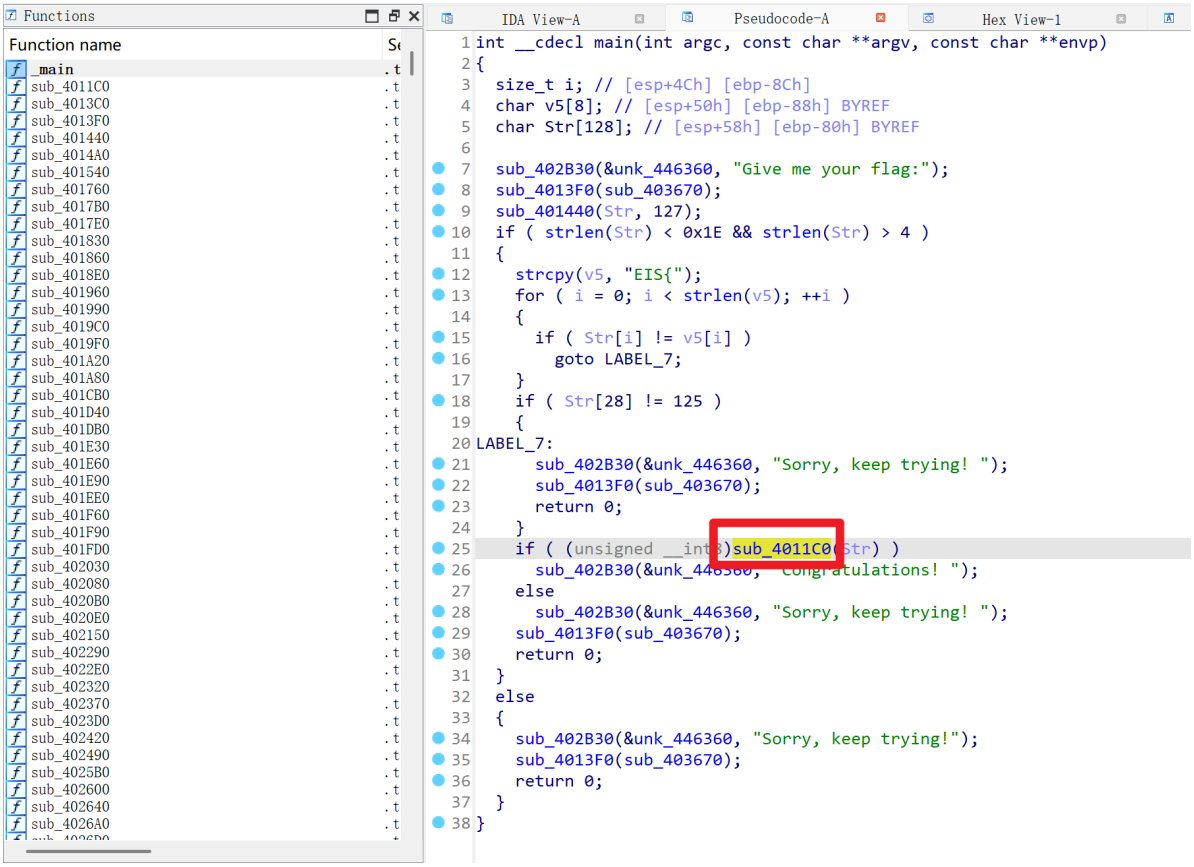


那么逻辑很清晰了，直接照着黑块后的代码进行逻辑的重写

```python
s = b"njCp1HJBPLVTxcMhUHDPwE7mPW"
flag = ""
for i, b in enumerate(s):
    if i % 2 == 0:
        flag += chr(b - 1)
    else:
        flag += chr(b + 1)
print(flag)
# flag{mkBqOIICOMUUwdLiTICQvF6nOX}
```

IgniteMe

IDA 打开后进入主函数，发现 flag 获取条件为 `sub_4011C0`



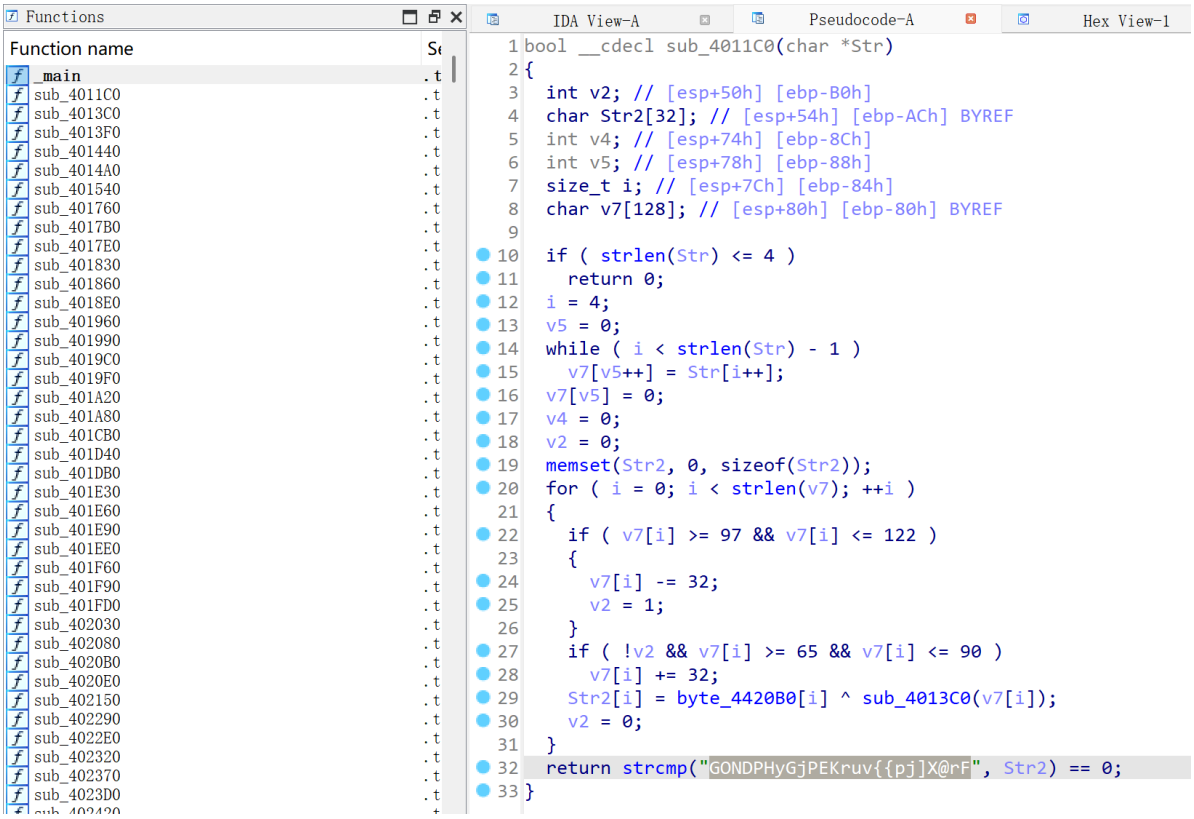
双击进入 `sub_4011C0` 后得到验证逻辑，首先要求输入大于 4

然后 i = 4 即从 flag 的第四位开始，进行了两步操作

1. 第 22 - 28 行，对 flag 进行大小写的互换
2. 第 29 行，逐位对字符进行了异或

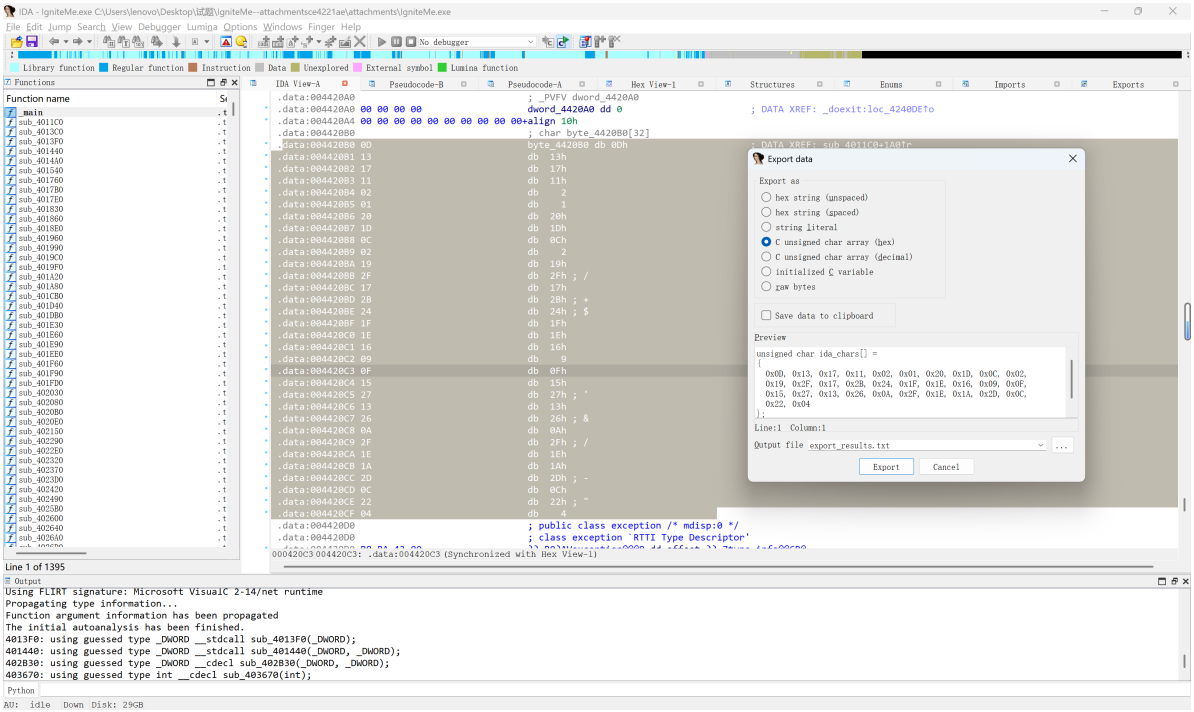
最后要求处理后的密文必须为

```
GONDPHyGjPEKruv{{pj]X@rF
```



先双击进入 `byte_4420B0` 提取所需的数据，选中所需数据后按 `Ctrl+E` 即可选择导出格式

```
[
  0x0D, 0x13, 0x17, 0x11, 0x02, 0x01, 0x20, 0x1D, 0x0C, 0x02,
  0x19, 0x2F, 0x17, 0x2B, 0x24, 0x1F, 0x1E, 0x16, 0x09, 0x0F,
  0x15, 0x27, 0x13, 0x26, 0x0A, 0x2F, 0x1E, 0x1A, 0x2D, 0x0C,
  0x22, 0x04
]
```



另外，双击进入 `sub_4013C0` 得到如下内容

```
int __cdecl sub_4013C0(int a1)
{
  return (a1 ^ 0x55) + 72;
}
```

那么整体逻辑已经理顺了，所需的数据也得到了，直接逆向操作即可

```
str1 = "GONDPHyGjPEKruv{{pj]X@rF"
str2 = [
  0x0D, 0x13, 0x17, 0x11, 0x02, 0x01, 0x20, 0x1D, 0x0C, 0x02,
  0x19, 0x2F, 0x17, 0x2B, 0x24, 0x1F, 0x1E, 0x16, 0x09, 0x0F,
  0x15, 0x27, 0x13, 0x26, 0x0A, 0x2F, 0x1E, 0x1A, 0x2D, 0x0C,
  0x22, 0x04
]

flag = ""

for i in range(len(str1)):
    # 异或
    val = ((ord(str1[i]) ^ str2[i]) - 72) ^ 0x55
    # 大小写字母互转
    if ord('a') <= val <= ord('z'):
        val = val - 32
    if ord('A') <= val <= ord('Z'):
        val = val + 32

    flag += chr(val)

print(flag)

# EIS{wadx_tdgk_aihc_ihkn_pjlm}
# 提交时答案开头是 EIS 还是 flag 请留意题目描述或两个都试试
```